

Tracking code in Git/GitHub

Research Tools Committee

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Center for Innovative Design & Analysis

colorado school of public health

Introduction

What is Git?

According to Git's website, "Git is a free and open source distributed version control system." So what does that mean? Let's break it down. First, Similarly to R, Git is available to anyone with a computer and access to the internet free of charge. Anyone can install and use it, and anyone can contribute to the project (if they have the technical know-how).

As for the second half, a version control system (VCS) is a tool that enables teams as small as a single person or as large as a multinational corporation to track changes in code over time and to integrate changes made by a decentralized team of developers.

For our purposes, Git enables us to create reproducible analysis code bases, where the full history of the analysis is available to ourselves and those which we choose to share it with.

What is GitHub?

Command Line Basics

In order to use Git from the command line (Terminal or Git Bash), a certain level of basic commands will be needed to navigate the files on your computer. Below is a list of commands that will help you navigate your directories and accomplish basic tasks using the command line.

- `ls` - List subdirectories and files in current directory
- `pwd` - Print working directory
- `cd` - Change directory (i.e., navigate to a different folder)
 - `cd MyFolder` - Move to the folder `MyFolder` located in current working directory
 - `cd Path/To/MyFolder` - Move to the folder `MyFolder` located at `Path/To/` inside the current working directory
 - `cd ..` - Move to parent directory (i.e. one folder back)
 - `cd /` - Move to root directory
 - `cd ~` - Move to home directory
- `mkdir` - Make directory
- `rm <filename>` - Remove file

Notes

Terminal / Git Bash

The instructions which follow attempt to be applicable to both MacOS and Windows platforms. However, some key differences do exist between the two operating systems. In most of these situations we have created separate instructions for Windows and MacOS. However, for sections where MacOS and Windows are functionally the same, only a single section is provided. In these sections, some language used may reference Terminal, and in those cases you should substitute Git Bash if you are using Windows operating system.

Main vs Master

Historically, the default branch in a new Git repository was named `master`. However, in 2020 a push to remove unnecessary references to slavery led GitHub and other companies to change the default branch name to `main`. The instructions which follow will assume the default branch name is `main` and will show you how to setup your local Git configuration to default to `main`.

However, due to the relative recency of this change, you may encounter repositories that have `master` as the primary branch. In such situations, all the instructions which follow will still be applicable, but you will need to substitute `main` for `master`.

Getting Started

Installing Git

MacOS

Open Terminal and run the command `git --version`. If you don't have Git installed already, you will be prompted to install. Follow the instructions provided in the Terminal or pop-up window to install Git.

Windows

Git and Git Bash come included as part of the [Git For Windows](#) package. Download and install Git For Windows like other Windows applications. Once downloaded find the included `.exe` file and open to execute Git Bash.

Creating GitHub Account

Go to <https://github.com/> and click **Sign Up** in the upper right hand corner. Follow the instructions to create an account, choosing an account name that is easily identifiable as belonging to you (i.e. "firstname-lastname" or something similar).

NOTE: If you already have a personal GitHub account, you may continue to use it for your work at CIDA **if** your account name is easily identifiable as belonging to you (i.e. if you have an account name like "firstname-lastname" or something similar).

Configure Git

To associate your local Git configuration with your name and Email, run the commands below in Terminal or Git Bash. Here, `your_email@cuanschultz.edu` should be substituted with the email associated with your GitHub account. If you are using a preexisting personal GitHub account, this may or may not end with `cuanschultz.edu`.

```
git config --global user.name "Firstname Lastname"
git config --global user.email "your_email@cuanschultz.edu"
```

Additionally, your local Git should be configured to make `main` the default branch name. To do so, run the following command in Terminal/GitBash:

```
git config --global -add init.defaultBranch main
```

Requesting CIDA GitHub Access

Following account creation, send an email containing your GitHub username to ryan <dot> peterson <at> cuanschutzh <dot> edu and max <dot> mcgrath <at> cuanschutzh <dot> edu to request access to CIDA's GitHub organization (please send email from your cuanschutzh.edu email address).

Setting Up SSH

The below instructions are up-to-date as of 10/05/22. Newer instructions along with additional troubleshooting may be available from [GitHub](#)

MacOS

1. Check for Existing SSH Keys
 - a. Open Terminal
 - b. Enter `ls -al ~/.ssh` to see if existing SSH keys are present by looking for the following filenames:
 - `id_rsa.pub`
 - `id_ecdsa.pub`
 - `id_ed25519`
 - c. If you see any of these files present, proceed to Step 3. Otherwise continue with Step 2.
2. Generate a new SSH key
 - a. Open Terminal
 - b. Enter `ssh-keygen -t ed25519 -C "your_email@example.com"` substituting in the email address associated with your GitHub account
 - c. When you're prompted to "Enter a file in which to save the key," press Enter. This accepts the default file location.
 - d. At the prompt, type a secure passphrase
 - e. Start the ssh-agent in the background by running `eval "$(ssh-agent -s)"`
 - f. Open the configuration file with `open ~/.ssh/config`
 - If the file doesn't exist create it with `touch ~/.ssh/config` then use the above command to open it

- g. Edit `~/.ssh/config` to contain the following lines:

```
::: {.cell}
```

```
Host *
  AddKeysToAgent yes
  UseKeychain yes
  IdentityFile ~/.ssh/id_ed25519
```

```
:::
```

- h. Add your SSH private key to the ssh-agent and store your password in the keychain by running `ssh-add --apple-use-keychain ~/.ssh/id_ed25519`

3. Add SSH key to GitHub account

- Copy the SSH public key to your clipboard with `pbcopy < ~/.ssh/id_ed25519.pub`
- Open GitHub in a web browser, log in
- Go to the upper right hand corner, click your profile photo, and select **Settings**
- Select **SSH and GPG keys** in the menu on the left
- Click green **New SSH key** button
- Enter a title for the SSH key in the **Title** field (use descriptive title like “CIDA MacBook Pro”)
- Select **Key type** as “Authentication Key”
- Paste your key into the **Key** field
- Click **Add SSH key**

4. Verify connection

- Test access to GitHub SSH with `ssh -T git@github.com`
- If you see the following message, verify that the fingerprint you see matches GitHub’s public key fingerprint ([link](#)). If it does, type **Yes**

```
::: {.cell}
```

```
```{.bash .cell-code}
```

```
> The authenticity of host 'github.com (IP ADDRESS)' cant be established.
> RSA key fingerprint is SHA256:nThbg6kXUpJWG17E1IGOCspRomTxdCARLviKw6E5SY8.
> Are you sure you want to continue connecting (yes/no)?
```

```
```
```

```
:::
```

- Verify that the resulting message contains your username. If you receive a “permission denied” message, see [Error: Permission denied \(publickey\)](#)

Windows

1. In Git Bash, type `ssh-keygen`
2. Press enter to save the SSH key to the default location
3. Enter a password to password protect your SSH key or leave blank and hit **Enter** to proceed without a password
4. You will receive a message saying your public key has been saved in `c/Users/username/.ssh/id_rsa.pub`, type `cat c/Users/username/.ssh/id_rsa.pub` to output the key
5. Copy and paste the key
6. Add SSH key to GitHub account
 - a. Open GitHub in a web browser, log in
 - b. Go to the upper right hand corner, click your profile photo, and select **Settings**
 - c. Select **SSH and GPG keys** in the menu on the left
 - d. Click green **New SSH key** button
 - e. Enter a title for the SSH key in the **Title** field (use descriptive title like “CIDA MacBook Pro”)
 - f. Select **Key type** as “Authentication Key”
 - g. Paste your key into the **Key** field
 - h. Click **Add SSH key**

Setting up SSO

Copy/distill instructions here: <https://docs.github.com/en/enterprise-cloud@latest/authentication/authenticating-with-saml-single-sign-on/about-authentication-with-saml-single-sign-on>

1. After being granted access to the CIDA-CSPH GitHub organization and setting up SSH, go to <https://github.com>, log in
2. In the upper-right corner of any page, click your profile photo, then click **Settings**
3. Select **SSH and GPG keys** in the menu on the left
4. To the right of the SSH key you’d like to authorize, click **Configure SSO**. If you don’t see **Configure SSO**, ensure that you have authenticated at least once with the CIDA-CSPH organization by visiting <https://github.com/orgs/CIDA-CSPH/>, accessing a non-public repository, and signing in with your CU login
5. You will see **CIDA-CSPH** listed. Click **Authorize** to the right.

Creating New Project

1. Create project folder on your local computer by calling `CIDAtools::create_project()` in R

2. Initialize git repository by navigating to folder (`cd Path/To/Folder`) then using the command `git init`
3. Stage file changes `git add .`
4. Create initial commit with `git commit -m "Initial commit"`
5. Create empty repository in CIDA-CSPH GitHub organization
 - a. In a web browser, navigate to <https://github.com/orgs/CIDA-CSPH/repositories>
 - b. Sign in using your CU login
 - c. Click the green **New repository** button
 - d. Enter the name of your repository
 - e. Do **not** add a template, README, .gitignore, or license file
 - f. Click **Create repository**
6. Add remote to local repository with `git remote add origin git@github.com:CIDA-CSPH/<your-repository>.git` (this SSH link can be copied from the empty GitHub remote repository you've just created).
7. Push work to GitHub with `git push origin main`

Cloning Existing Project

1. Navigate to GitHub remote repository
2. Click green "Code " button
3. Click "SSH" then copy link
4. Clone repository with `git clone git@github.com:CIDA-CSPH/<your-repository>.git`

Regular Workflow

1. Add and commit changes

```
cd Path/To/Directory
git add .
git commit -m "My commit message"
```

2. Fetch and merge any changes (see Handling Merge Conflicts below)

```
git fetch origin main
git merge origin/main
## Fix any merge conflicts
git commit -m "Brief description of changes (<=50 characters)"
```

3. Push all new commits

```
git push origin main
```

Handling Merge conflicts

In the case that another user has modified and committed changes to a file that you have modified in one of your recent commits, when you run `git pull` you may be notified that you have a merge conflict and need to resolve those conflicts before you can push your changes to GitHub. To do so:

1. After running `git merge origin/main` and being notified that you have merge conflicts, run `git status` to see which files have conflicts (they will be listed with **both modified:** in front of them)
 - a. Note: On newer version of Git you may receive an error saying “You have divergent branches and need to specify how to reconcile them”
 - b. In this case, you can add an indicator to pull without rebasing with `git merge --no-rebase origin/main`
2. Open those files in a text editor (RStudio, Vim, textEdit, Notepad++, etc.)
3. Here, you will see some sections of code with:

```
<<<<<<< HEAD
# Version 1 of Code
=====
# Version 2 of Code
>>>>>>> commit_hash
```

4. Here, the section of code between `<<<<<<< HEAD` and `=====` is the local version of the code, while the code between `=====` and `>>>>>>> commit_hash` is the version of the code pulled from the remote (i.e., GitHub)
5. To fix the merge conflict, simply select which version of the code you want to keep and delete the other version along with `<<<<<<< HEAD` and `=====` and `>>>>>>> commit_hash`. Note that you may also need to mix and match between the two sections of code, but always delete the conflict markers.
6. Save your changes
7. Return to Terminal/Git Bash and add commit changes

```
git add .
git commit
```


Additional Git Topics

Branching

Stashing changes

Helpful Git Commands

FAQ

Do we use GitHub? Or GitLab? Didn't we just switch to GitHub?

Either are currently viable options. We are currently transitioning from GitLab to GitHub, so if you are picking this up for the first time, I suggest using GitHub (and will call both "GH" from here on for simplicity).

Where does my main analysis code "live"? When I'm committing and pushing to GitHub/Lab am I just making backups of my work that has its "home" on the P: drive? Or should the code "live" on the cloud in GitHub/GitLab, and every time I want to work with it, I should pull it down?

Your code will live on GH. Repos from GH can be cloned to wherever you want; locally, the P drive, etc. So when you need to run your code, your machine can read it from a local or network location. It's best practice to pull from GH before you need to run anything for a project in case someone's changed (if nothing has changed it will say "Your repo is up to date").

Should I even then have a version of my code on the P: drive at all?

It's up to you

If I do, and I pull from GitHub/Lab, am I replacing the code on the P: drive?

Yes, but only if something's changed on the GitHub server. You can also easily "checkout" earlier versions of the repository from previous points in time, so the older code is never lost.

One advantage that people talk about a lot for using Git is that it can "merge" files from multiple people. How does this actually work?. What's to prevent person A from pulling the syntax and making edits, person B pulling the syntax and making edits, and then both people pushing up to a Git server? Wouldn't there then be 2 versions of the file, neither of them fully correct? How is this an upgrade from not having Git?

Git will force the person who pushes second to fix any merge issues in the code before pushing back to GitHub in this case. There are also more tools available such as branching if this issue comes up often for a particular project.

I don't really use that much version control now anyways, ad-hoc or otherwise. Is this bad? Have I just not been exposed to projects that change enough to merit it?

CIDA requires code be tracked in Git and GitHub. Your future self will thank you.

Often in my code I include additional “justification” code that’s commented out. For example, if I merge two datasets on ID, I also check the dimensions of the new merged dataset to make sure it is what I expected. Or, if I write a function, I also hard-code a version of the function to make sure the function actually does what it’s supposed to. These examples are a bit trivial, but I think they illustrate well the sort of “double-checking” that I sometimes include code for. Is this the sort of thing you think is valuable to include in code as a comment? Or does that bother some people, and should it be taken out of the syntax after the double-check has been completed?

I think what you are referring to is “unit testing”, which is in fact good practice. I would put all new functions you create in a separate `helper-functions.R` file which can be scripted from your Rmd, and you can also have your (self-contained) unit tests there as well.

Isn't there a CIDA element to the GH platform that we need to use, that is more “secure” or something? Or are we only supposed to use our personal GH accounts for all our work?

CIDA has a GitHub “organization” that is part of the CU GitHub Enterprise account. Membership to this organization can be merged with your personal GitHub account or a new GitHub account. All university- (CIDA-) related repos should live under this umbrella. The main page is <https://github.com/CIDA-CSPH>; you will need an account invitation; please contact a member of the Research Tools committee for an invitation.

Git Kraken (Legacy)

This section will walk through how to set up GitKraken to interface with the CIDA GitHub, allowing you to easily push/pull projects between your working directory and GitHub.

First, you will have to download GitKraken at <https://www.gitkraken.com/download>. Once installed, you will sign in with the GitHub account associated with the CIDA GitHub. This will take you to the SSO associated with the CIDA GitHub, which you will then authorize and sign onto.

Welcome to GitKraken



Using a hosting service?



GitHub



GitLab



Bitbucket

If you've already connected with GitHub, sign in below.



Sign in with GitHub

Open GitKraken?

https://api.gitkraken.com wants to open this application.

☐ Always allow api.gitkraken.com to open links of this type in the associated app

Open GitKraken

Cancel

Success!

Congratulations, your connection was successful!

GitHub and **GitKraken** are good to go!



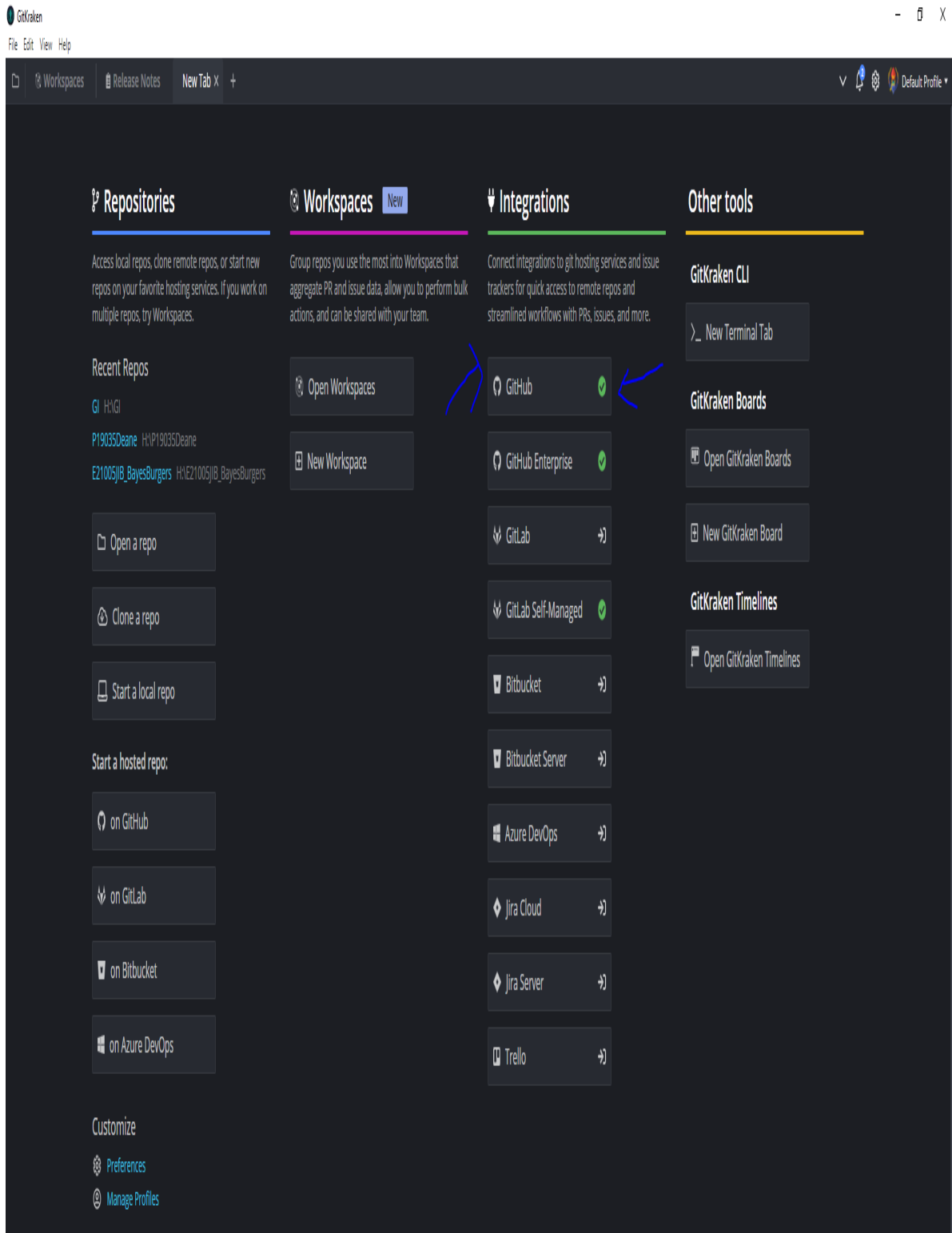
Didn't work?

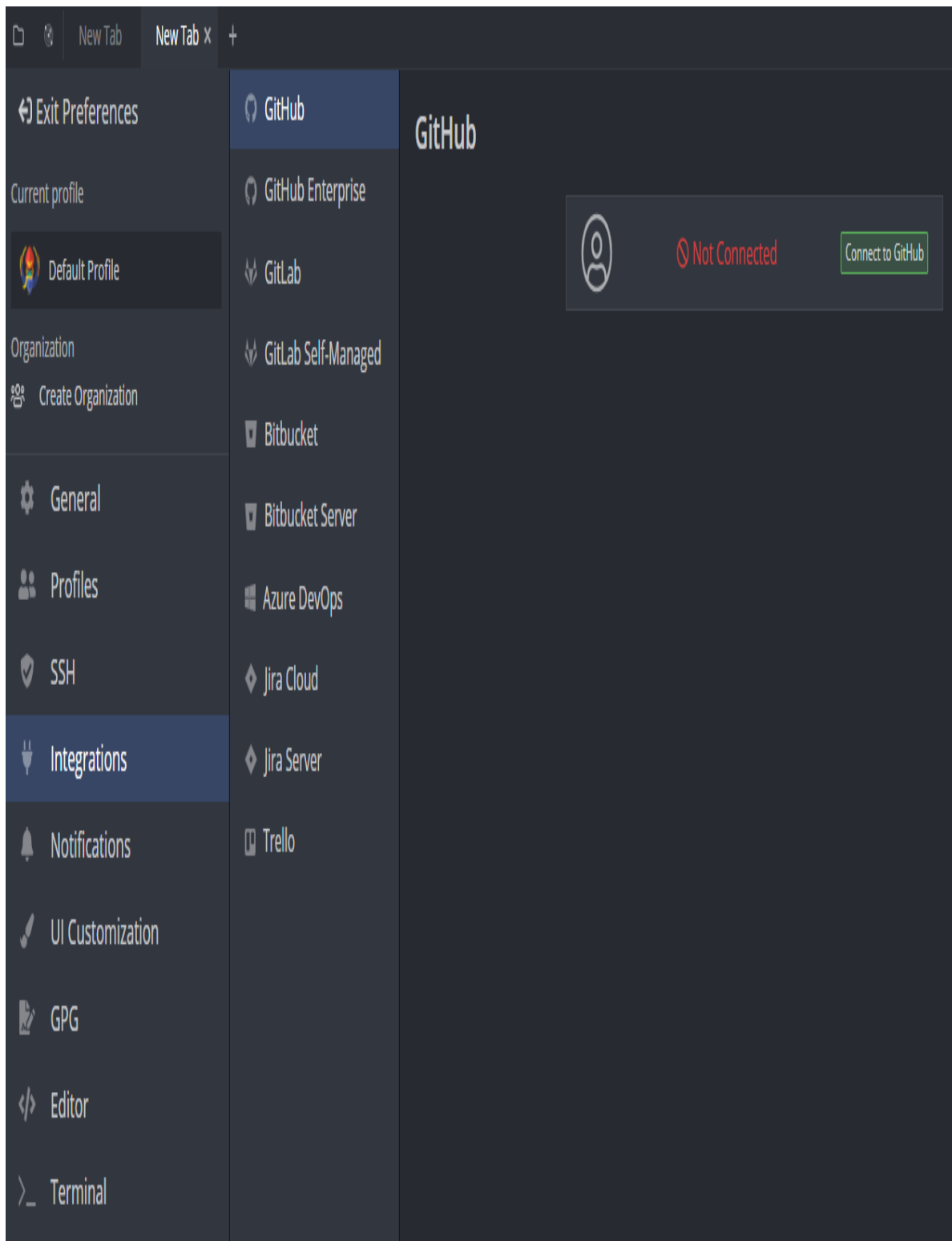
Copy the token and paste it into GitKraken.

```
eJwtzE2PgjAUheH/0jULEJDS5TgZ0cTEhRmT2TT9uGil0OYWBo3xv9sYdicned4nEUpBCCfXwUAYoaqmVaFpl
```

Copy Token

Once you have logged into GitKraken with your GitHub account, you will need to connect it to GitHub. The default GitKraken screen has a panel called “Integrations”: click on GitHub. From here, there are two possibilities: if you click on “Connect to GitHub” it may take you to the SSO again and you can log in that way (slightly repetitive). Or, if prompted to use the OAuth or Personal Access token, you can do the following: go to your “Settings” tab on GitHub, click on “Developer Settings”, and then click on “Personal access tokens” to generate a token for 90 days, making sure to select “repo” (along with other scopes as needed).





GitHub

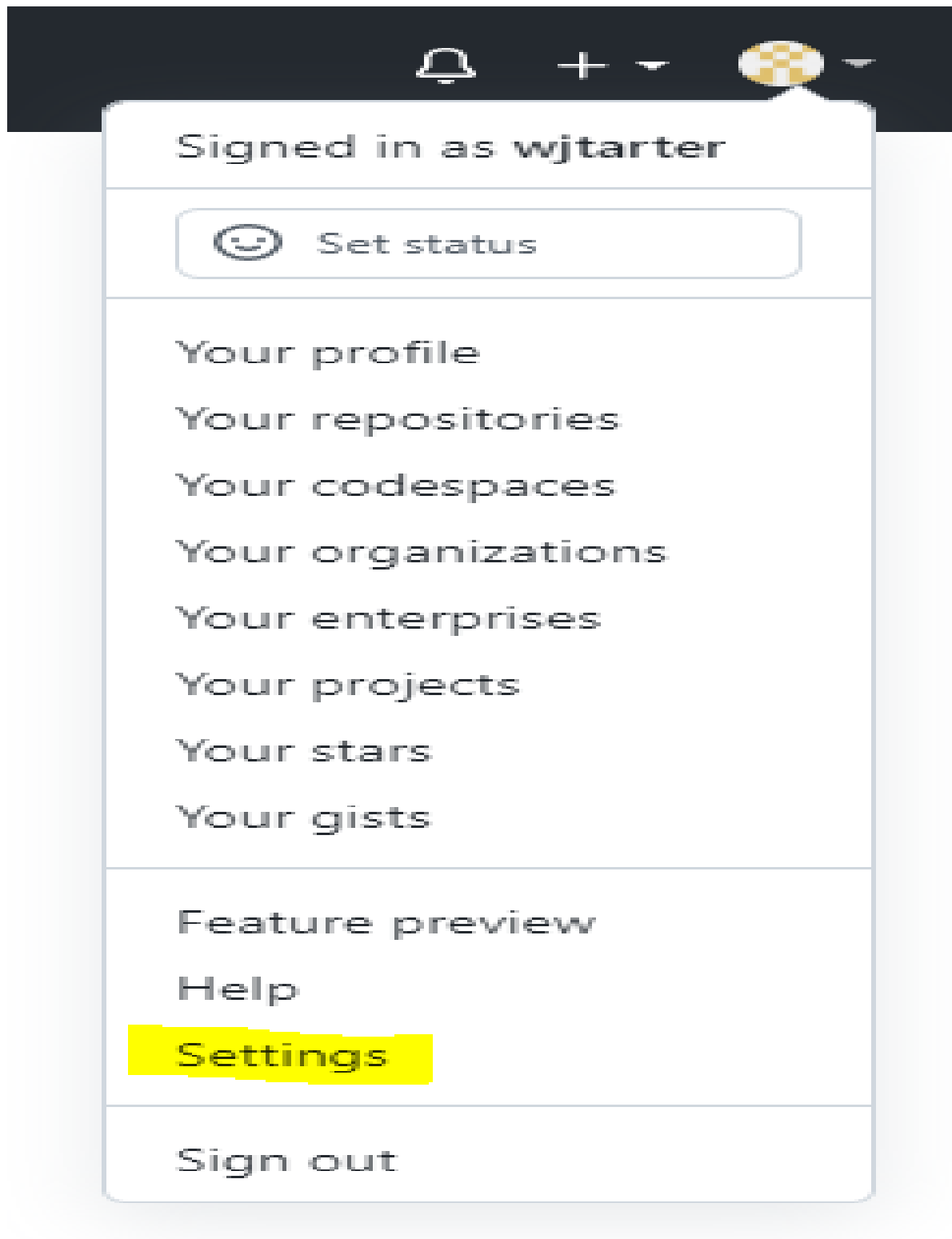


Connecting...

Connect to GitHub

OAuth Token

Save Token



-  Public profile
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-  Accessibility
-  Notifications

Access

-  Billing and plans
-  Emails
-  Password and authentication
-  SSH and GPG keys
-  Organizations
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Code, planning, and automation

-  Repositories
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Security

-  Code security and analysis

Integrations

-  Applications
-  Scheduled reminders

Archives

-  Security log
-  Sponsorship log

 Developer settings

GitHub Apps

OAuth Apps

Personal access tokens

New personal access token

Personal access tokens function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

CIDA Git

What's this token for?

Expiration *

90 days

The token will expire on Sun, Dec 18 2022

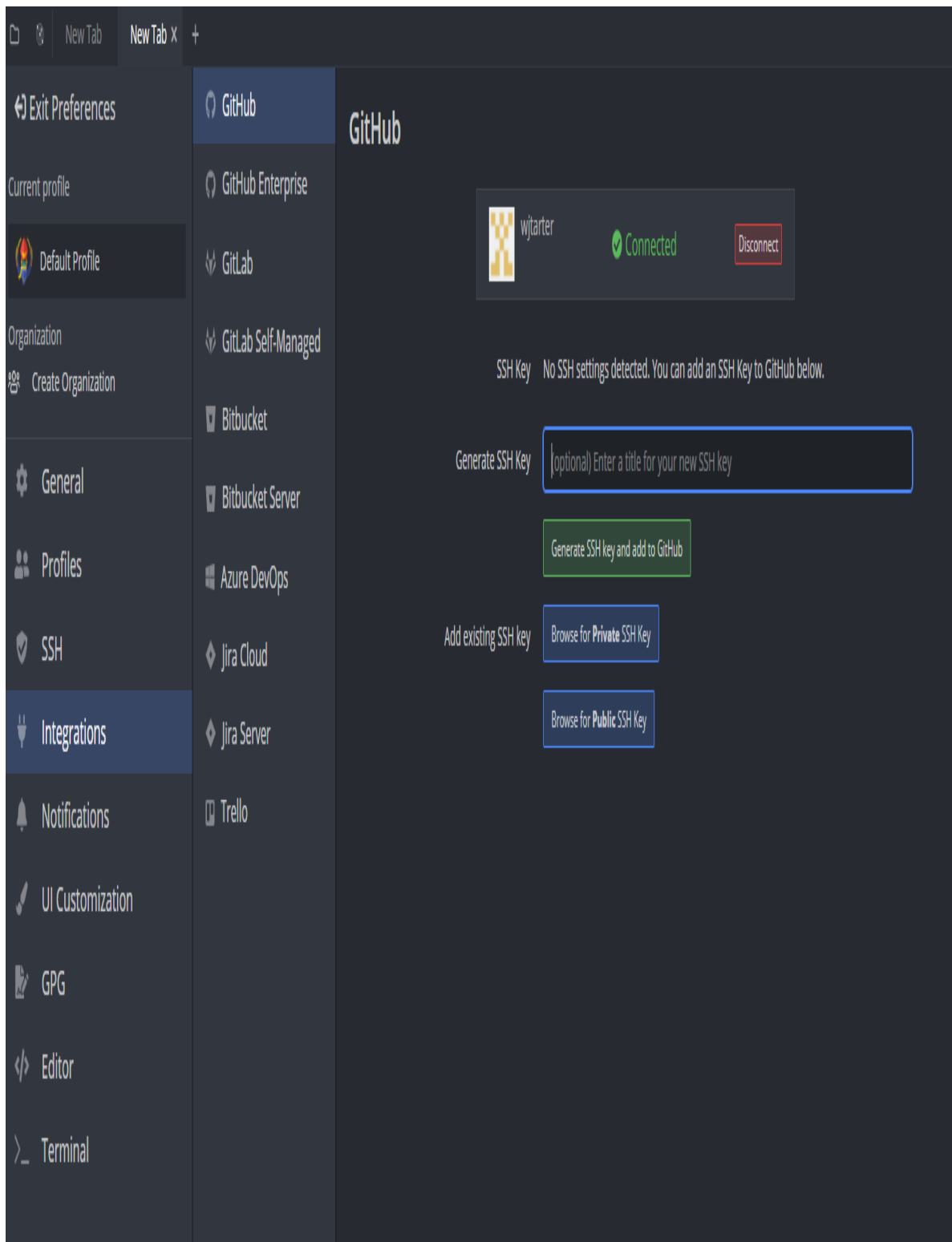
Select scopes

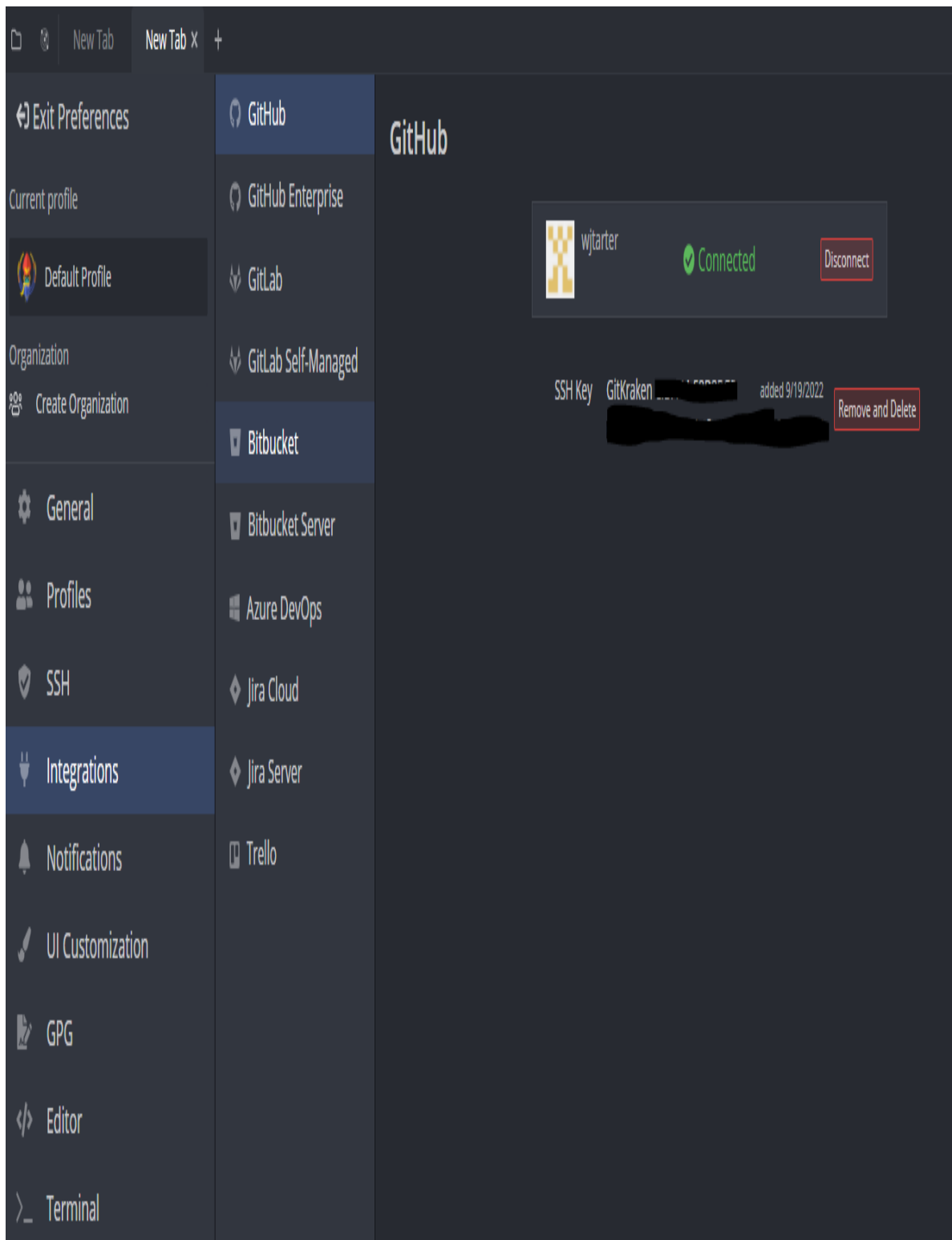
Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

| | |
|---|---|
| ☒ repo | Full control of private repositories |
| ☒ repo:status | Access commit status |
| ☒ repo:deployment | Access deployment status |
| ☒ public_repo | Access public repositories |
| ☒ repo:invite | Access repository invitations |
| ☒ security_events | Read and write security events |
| ☐ workflow | Update GitHub Action workflows |
| ☐ write:packages | Upload packages to GitHub Package Registry |
| ☐ read:packages | Download packages from GitHub Package Registry |
| ☐ delete:packages | Delete packages from GitHub Package Registry |
| ☐ admin:org | Full control of orgs and teams, read and write org projects |
| ☐ write:org | Read and write org and team membership, read and write org projects |
| ☐ read:org | Read org and team membership, read org projects |
| ☐ manage_runners:org | Manage org runners and runner groups |
| ☐ admin:public_key | Full control of user public keys |
| ☐ write:public_key | Write user public keys |
| ☐ read:public_key | Read user public keys |
| ☐ admin:repo_hook | Full control of repository hooks |
| ☐ write:repo_hook | Write repository hooks |
| ☐ read:repo_hook | Read repository hooks |
| ☐ admin:org_hook | Full control of organization hooks |

Ideally, you should be able to automatically connect to GitHub without generating a personal access token, but GitKraken is moderately cursed.

Once you have successfully connected to GitHub, you will then need to generate an SSH key and add to GitHub. To do this, simply click the “Generate SSH key and add to GitHub” button (optionally adding a title to the SSH key if you wish). If this step is successful, you will receive a notification within GitKraken on the lower left, and your screen will now display the SSH key within GitKraken and under the Settings -> SSH and GPG keys tab on your GitHub:





 **wjtarter**
Your personal account

[Go to your profile](#)[Public profile](#)[Account](#)[Appearance](#)[Accessibility](#)[Notifications](#)[Access](#)[Billing and plans](#)[Emails](#)[Password and authentication](#)[SSH and GPG keys](#)[Organizations](#)[Enterprises](#)[Moderation](#)[Code, planning, and automation](#)[Repositories](#)[Packages](#)[GitHub Copilot](#)[Pages](#)[Saved replies](#)

SSH keys

[New SSH key](#)

This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.

Authentication Keys

 **GitKraken** [REDACTED]
[REDACTED]
SSH Added on Sep 19, 2022 by GitKraken
Never used — Read/write

[Configure SSO](#) [Delete](#)

Check out our guide to [generating SSH keys](#) or [troubleshoot common SSH problems](#).

GPG keys

[New GPG key](#)

There are no GPG keys associated with your account.

Learn how to [generate a GPG key](#) and [add it to your account](#).

Vigilant mode

☐ Flag unsigned commits as unverified

This will include any commit attributed to your account but not signed with your GPG or S/MIME key.

Note that this will include your existing unsigned commits.

[Learn about vigilant mode.](#)

Congratulations, you should be successfully linked to the CIDA GitHub!

From here, you can open, clone, and initiate repositories from the CIDA Github using the, you guessed it, Open a repo, Clone a repo, and Start a local repo options on the home screen:

Repositories

Access local repos, clone remote repos, or start new repos on your favorite hosting services. If you work on multiple repos, try Workspaces.

Recent Repos

GI H:GI

P19035Deane H:P19035Deane

E21005JB_BayesBurgers H:E21005JB_BayesBurgers

Open a repo

Clone a repo

Start a local repo

Start a hosted repo:

on GitHub

on GitLab

on Bitbucket

on Azure DevOps

Customize

Preferences

Manage Profiles

Additional Resources

Watch the GitKraken Client intro video

Join the GitKraken Slack community

Read the GitKraken Client Documentation

Browse or Submit Feature Requests

Sign up for the GitKraken Newsletter

Provide Feedback

Get Support

Workspaces New

Group repos you use the most into Workspaces that aggregate PR and issue data, allow you to perform bulk actions, and can be shared with your team.

Open Workspaces

New Workspace

Integrations

Connect integrations to git hosting services and issue trackers for quick access to remote repos and streamlined workflows with PRs, issues, and more.

GitHub

GitHub Enterprise

GitLab

GitLab Self-Managed

Bitbucket

Bitbucket Server

Azure DevOps

Jira Cloud

Jira Server

Trello

Other tools

GitKraken CLI

New Terminal Tab

GitKraken Boards

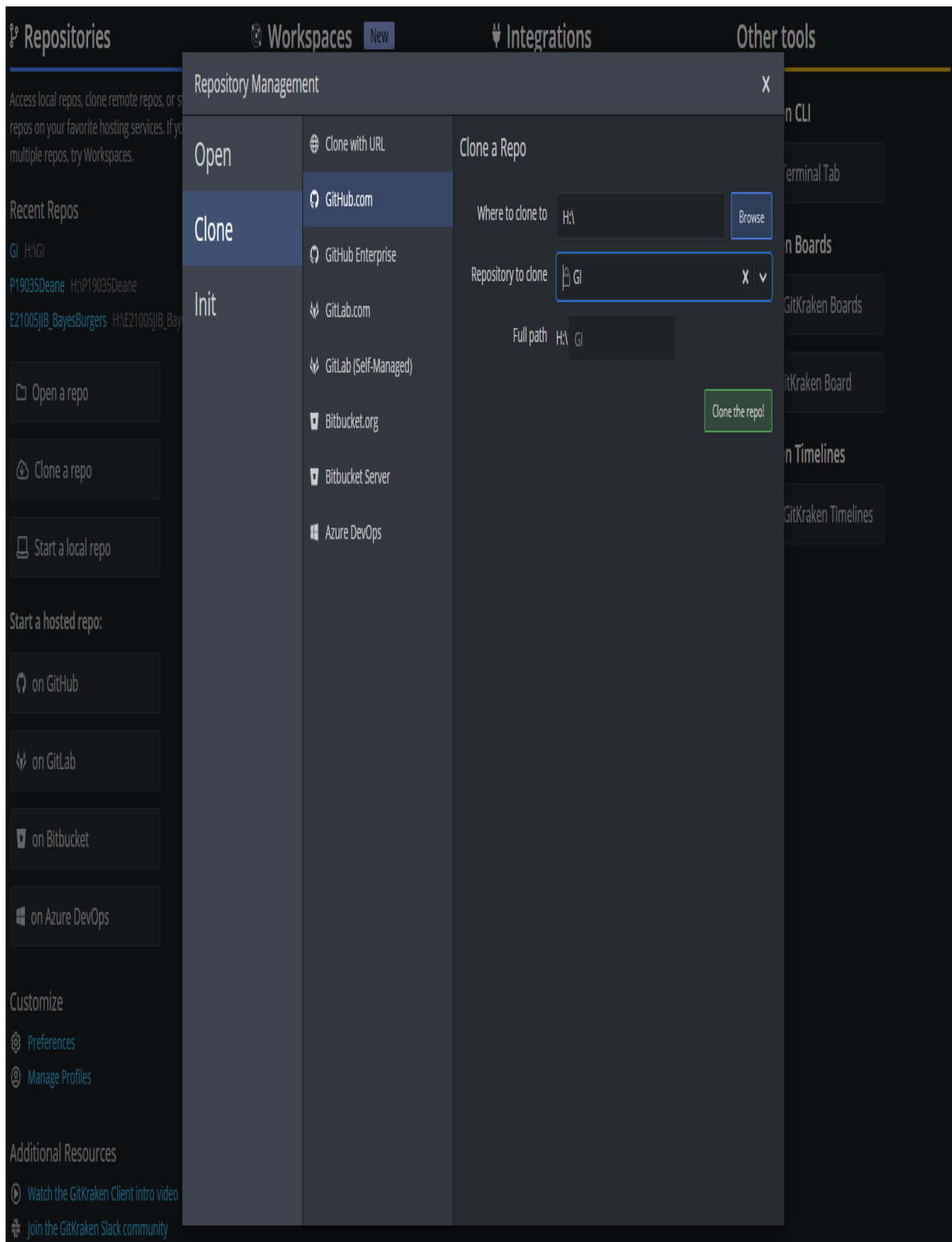
Open GitKraken Boards

New GitKraken Board

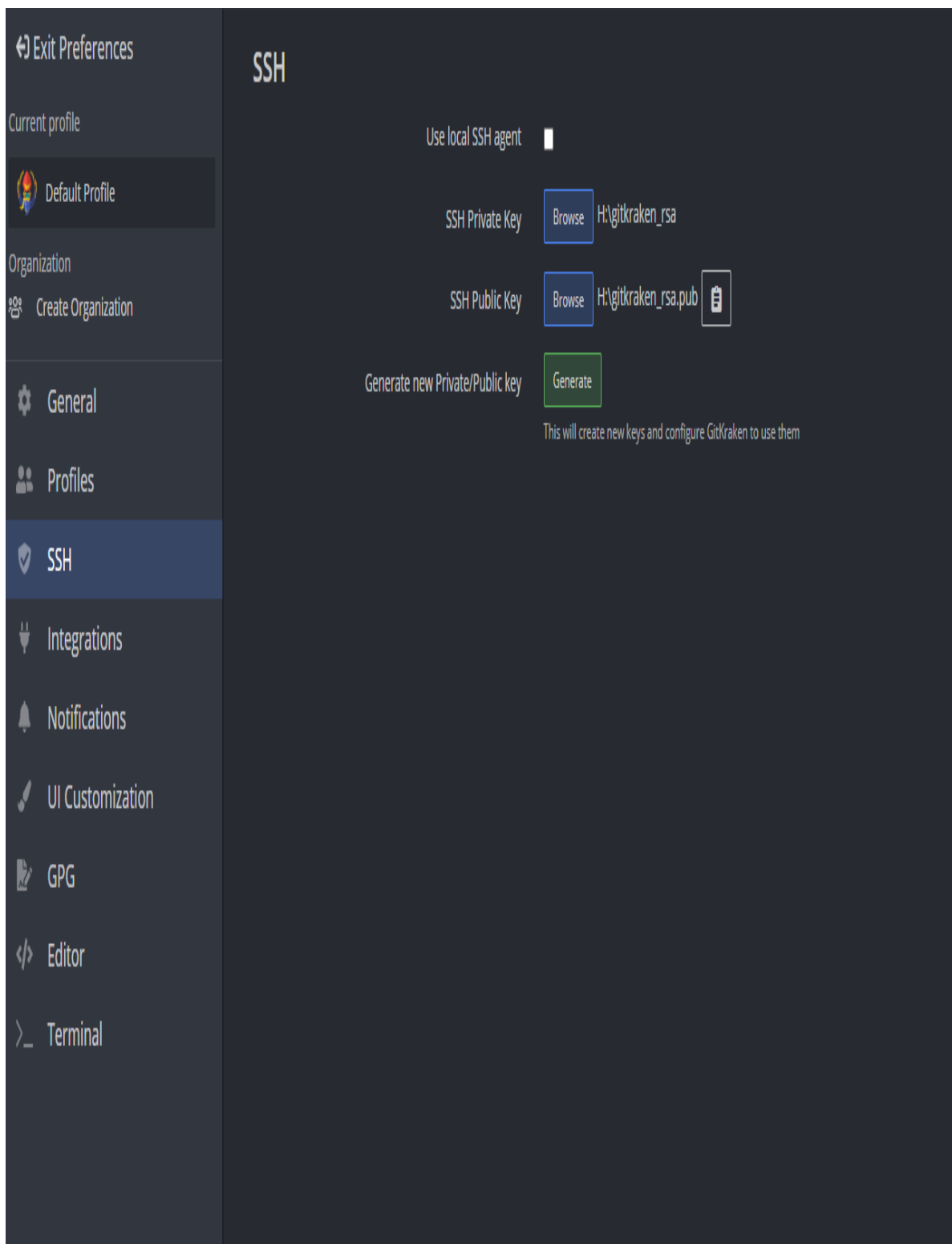
GitKraken Timelines

Open GitKraken Timelines

If you are going to clone a repo, just make sure you are on the GitHub.com section, and that you select the CIDA repository to clone, as well as where you would like to clone it to (H: or P: drives, or elsewhere).



First, however, you will need to generate another SSH key on GitKraken to add to your GitHub (at least this is the only way that I can figure). Go to Preferences -> SSH and then generate a new Private/Public Key, or copy the SSH Public Key if you already have one:



Then return to your GitHub account, and click on SSH and GPG keys under the settings, click Add new key, and then copy the SSH key into the Key box. You will then need to Authorize it with the SSO. After that, you should be able to clone, open, and initialize your repos!

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 **SSH and GPG keys**

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SSH keys / Add new

Title

Key type

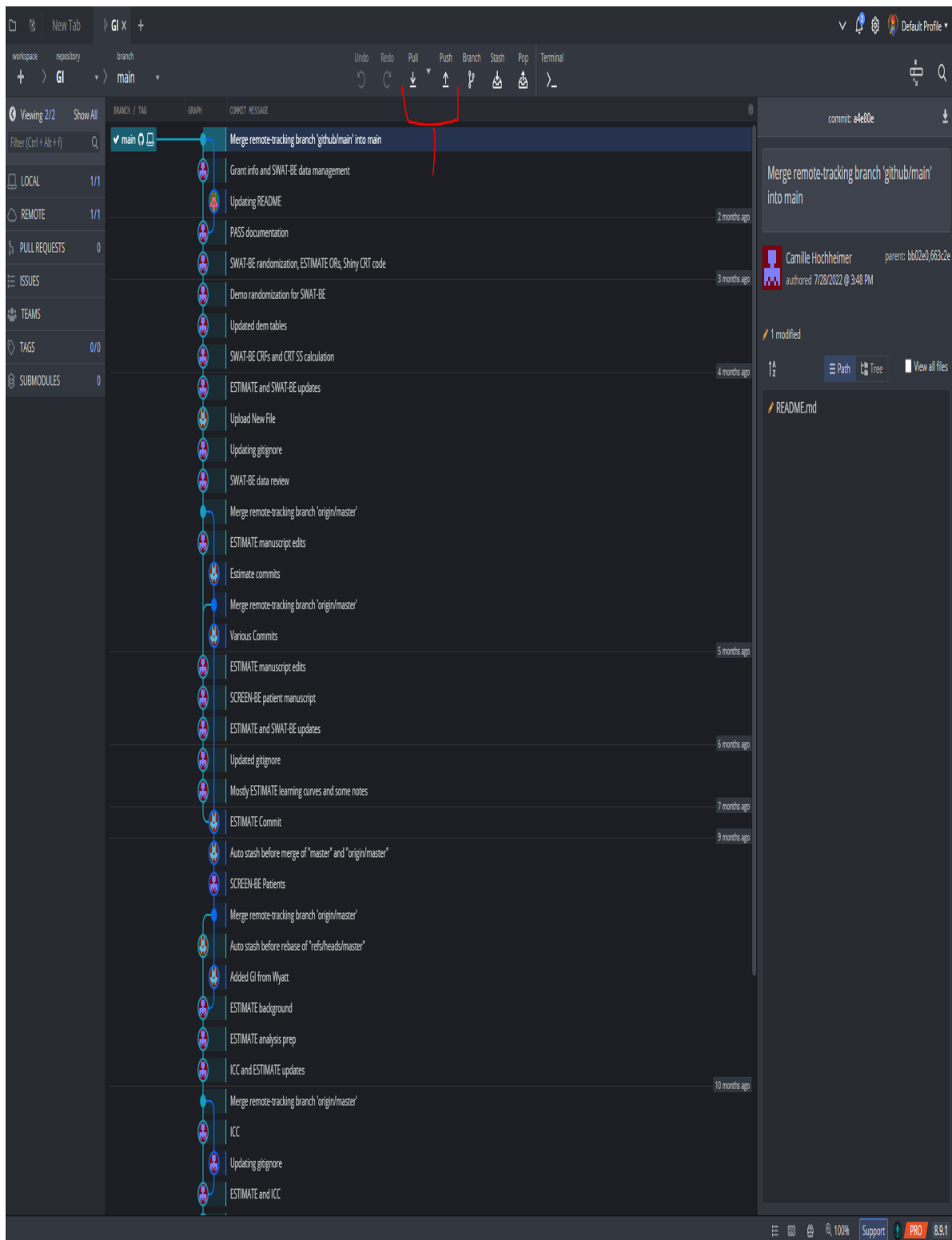
Authentication Key 

Key

Begins with 'ssh-rsa', 'ecdsa-sha2-nistp256', 'ecdsa-sha2-nistp384', 'ecdsa-sha2-nistp521', 'ssh-ed25519', 'sk-ecdsa-sha2-nistp256@openssh.com', or 'sk-ssh-ed25519@openssh.com'

[Add SSH key](#)

Once you have cloned a repo, you can open it to push/pull any changes. This is done pretty simply via the push or pull commands on the top after opening the repo.



For more details on how to manage pull, push, branching, and other features, visit <https://help.gitkraken.com/gitkraken-client/github-gitkraken-client/>

Center for Innovative Design & Analysis
colorado school of public health